

SmartInterview: Master Quick-Revision Cheatsheet & Viva Defense Card

Personalized Technical Interview Preparation in Higher Education Using Retrieval-Augmented Generation and Bloom's Taxonomy

(SmartInterview — G-1392)

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1. THE 30-SECOND ELEVATOR PITCH (MEMORIZE VERBATIM)

"SmartInterview is an AI-powered, adaptive technical mock interview platform that prepares engineering students for high-stakes recruitment interviews. Traditional preparation relies on static question banks or ungrounded conversational AI bots that hallucinate and grade subjectively. SmartInterview solves this through a decoupled 4-tier architecture: it parses resumes and job descriptions using PyMuPDF to extract skill gaps; grounds technical questions in a curated 402-chunk knowledge base in ChromaDB using RAG; drives interview progression through a deterministic state machine based on Bloom's Revised Taxonomy; and evaluates answers via an objective 5-signal formula combining LLM technical reasoning with dense vector cosine similarity. With real-time neural voice interaction powered by Groq LPU and Microsoft Edge-TTS, the system runs with sub-second latency at zero infrastructure cost."

2. MASTER SYSTEM NUMBERS & ARCHITECTURAL SPECS

Metric / Dimension	Exact System Value	Why This Number Matters
Knowledge Base Chunks	402 Chunks	Curated across 8 computer science domains in ChromaDB (technical_kb).

Technical Domains	8 Domains	cn (33), dbms (45), design-patterns (105), dsa (63), ml-dl (35), oop (33), os (35), system-design (53) — 50 concepts, 402 chunks.
Embedding Dimensions	384 Dimensions	Produced by sentence-transformers/all-MiniLM-L6-v2 locally on CPU.
Embedding Parameters	22.7 Million	Lightweight enough to vectorize sentences in under 15ms on commodity hardware.
Inference Generation	>500 tokens/sec	Groq Cloud LPU running Tensor Streaming Processors (TSP).
Time-to-First-Token	<180 ms	Delivers instantaneous conversational speech cadence during interviews.
Cognitive Levels	6 Bloom Tiers	<i>Remember $\rightarrow$ Understand $\rightarrow$ Apply $\rightarrow$ Analyze $\rightarrow$ Evaluate $\rightarrow$ Create.</i>
Scoring Signals	5 Signals	Technical (30%), Completeness (20%), Relevance (20%), Semantic (15%), Concepts (15%).
Relational DB Tables	7 MySQL Tables	users, interview_sessions, interview_questions, session_feedback, syllabus_catalog, user_progress, token_blacklist.
Chunking Parameters	300–500 words	50-word sliding window overlap to prevent contextual boundary truncation.
Voice Synthesis Model	en-US-ChristopherNeural	High-clarity professional male technical recruiter voice profile via Edge-TTS.
Speech-to-Text Model	Whisper-large-v3	Conditioned with technical vocabulary prompt biasing for CS terminology.

3. THE 4-TIER ARCHITECTURE AT A GLANCE

1. PRESENTATION TIER: React 18/19 SPA + Vite 6 + Tailwind CSS v4 - Resume & JD Upload UI - Skill Gap Matrix - Canvas Audio Waveform - Real-Time Interview Room 5-Axis Radar Chart Performance Dashboard
2. APPLICATION TIER: FastAPI 0.115 ASGI + Python 3.10+ + Uvicorn - PyMuPDF C-level resume parser - Deterministic Bloom state machine 5-signal rubric evaluator - In-memory neural voice streaming
3. PERSISTENCE TIER: MySQL 8.0 (InnoDB) + ChromaDB 0.6.3 - MySQL: Strict ACID transactions, 7 relational tables, JSON state column - ChromaDB: Local HNSW cosine vector index, 402 CS chunks, 0ms network
4. INFERENCE & VOICE TIER: Groq LPU + Microsoft Edge-TTS - Groq LPU: GPT-OSS-120B / Llama-3.3-70B (>500 t/s) + Whisper-large-v3 - Edge-TTS: In-memory async chunked WebSocket audio streaming (Zero cost)

4. FORMULA REFERENCE CARD

1. Hybrid 5-Factor Answer Scoring Equation

$$\text{Score} = 0.30 \cdot T + 0.20 \cdot C + 0.20 \cdot R + 0.15 \cdot S + 0.15 \cdot K$$

- \$T\$ (Technical Accuracy - 30%):** Factual precision, correct syntax, absence of misconceptions (LLM graded).
- \$C\$ (Completeness - 20%):** Depth of explanation, edge-case coverage (LLM graded).
- \$R\$ (Relevance - 20%):** Direct responsiveness to prompt, zero topic drift (LLM graded).
- \$S\$ (Semantic Similarity - 15%):** Cosine distance between answer vector and top-3 RAG chunks (SentenceTransformers).
- \$K\$ (Concept Coverage - 15%):** Ratio of essential domain keyphrases present in candidate answer.

2. Dense Vector Cosine Similarity

$$\text{Cosine Similarity}(\mathbf{a}, \mathbf{r}) = \frac{\mathbf{a} \cdot \mathbf{r}}{\|\mathbf{a}\|_2 \|\mathbf{r}\|_2} = \sum_{i=1}^{384} a_i r_i \quad (\text{since vectors are } L_2 \text{ normalized})$$

3. Adaptive Engine Bloom Progression Rules

- Overall Score ≥ 75 :** Advance Bloom tier ($\text{Remember} \rightarrow \text{Understand} \rightarrow \text{Apply} \rightarrow \text{Analyze} \rightarrow \text{Evaluate} \rightarrow \text{Create}$). If already at **Create**, increase difficulty to **hard**.

- **Overall Score \$50 - 74\$:** Maintain Bloom tier. Rotate `question_type` (conceptual $\rightarrow$ practical $\rightarrow$ scenario) to test horizontal mastery.
- **Overall Score < 50 \$:** Regress Bloom tier ($\text{Create} \rightarrow \dots \rightarrow \text{Remember}$) to scaffold foundational gaps. If already at `Remember`, set difficulty to `easy`.

4. Priority Skill Rotation Heuristic

$$P(s) = \frac{1}{N_s + 1} \cdot \left(1.0 - \frac{\bar{S}_s}{100}\right)$$

- N_s : Number of times skill s has been evaluated.
- $\bar{S}_s$: Candidate's running average score on skill s .
- Prioritizes skills with fewer evaluations and lower performance.

5. "WHY THIS, NOT THAT?" SPEED DEFENSE MATRIX

Component	Selected Tool	Rejected Alternatives	Instant 1-Sentence Defense
Backend	FastAPI	Flask, Django, Node.js	Native <code>async/await</code> for audio streaming, automatic Pydantic v2 validation, and zero-overhead Python AI runtime.
LLM Inference	Groq LPU	OpenAI GPT-4o, Claude 3.5, Ollama	Groq's Tensor Streaming Processor delivers >500 tokens/sec (<180ms TTFT) for real-time oral dialogue at zero cost.
Vector DB	ChromaDB	Pinecone, Milvus, pgvector	Embedded in-process with 0ms network latency, native metadata filtering, and zero subscription cost.
Embeddings	all-MiniLM-L6-v2	OpenAI text-embedding-3, BGE-Large	384 dimensions compute in <15ms on CPU with zero cloud fees, perfectly matching semantic sentence matching.
PDF Ingestion	PyMuPDF (<code>fitz</code>)	pdfplumber, pypdf, Apache Tika	High-performance C-engine extracts PDF glyph streams in 12ms without requiring a heavy Java runtime.

Voice Output	Microsoft Edge-TTS	Web Speech API, ElevenLabs	Streams natural neural speech in-memory via WebSockets, avoiding robotic browser synthesizers and high API fees.
Voice Input	Groq Whisper-v3	Browser Web Speech Recognition	Universal browser compatibility and superior technical jargon spelling via vocabulary prompt biasing.
Database	MySQL 8.0	MongoDB, PostgreSQL, DynamoDB	Strict ACID transactions, native JSON columns for AdaptiveState, and institutional curriculum compliance.
Framework	Bloom's Taxonomy	1–10 Difficulty Numbers, LeetCode	Measures qualitative cognitive depth (Analyze/Evaluate/Create) rather than arbitrary question obscurity.
Scoring	Hybrid 5-Factor	Single-Prompt Zero-Shot LLM	Decouples grading into 5 orthogonal signals; immune to confident buzzword bluffing (\$r=0.89\$ human correlation).
State Engine	Deterministic Python	Autonomous LLM Agent Loop	Eliminates stochastic mood drift, token waste, and hallucination; guarantees mathematical progression rules.

6. AI VS. DETERMINISTIC LOGIC: THE CRITICAL DISTINCTION

Architectural Subsystem	Deterministic Software Logic (Python/MySQL/Chroma)	Generative AI / Deep Learning (Groq/Whisper/Edge-TTS)
Question Formulation	Computes query vector, filters domain chunks, enforces Bloom prompt	Synthesizes natural, varied question text adhering strictly to context

Cognitive Progression	Checks thresholds ($\geq 75, 50-74, <50$), advances/regresses Bloom state	ZERO control (LLM is never allowed to decide what level comes next)
Skill Selection	Evaluates priority formula $P(s)$, selects least-tested skill	ZERO control
Answer Scoring	Computes Cosine Similarity (SS), keyword coverage ratio (SK), and final sum	Grades Technical Accuracy (TS), Completeness (CS), and Relevance (RS)
Audio Processing	AudioContext frequency analysis, canvas waveform rendering	Whisper-large-v3 neural STT + Edge-TTS neural speech synthesis
Session Persistence	Commits relational ACID transactions, serializes AdaptiveState JSON	ZERO control

7. TOP 25 VIVA QUESTIONS & CRISP KILLER ANSWERS

1. What is SmartInterview?

An AI-powered adaptive technical mock interview platform that grounds questions in a 402-chunk CS knowledge base via RAG, adapts difficulty via Bloom's Taxonomy, and evaluates answers using a 5-signal hybrid formula.

2. Why use RAG instead of asking the LLM directly?

Ungrounded LLMs hallucinate technical facts and drift off-topic. RAG anchors question generation in verified computer science literature.

3. What is the embedding model and where does it execute?

`all-MiniLM-L6-v2` (384 dimensions, 22.7M parameters) running locally on the server CPU in under 15ms.

4. Why ChromaDB over cloud vector databases like Pinecone?

ChromaDB runs in-process with 0ms network latency, requires no cloud subscriptions, and supports native dictionary metadata filtering.

5. What is the mathematical formula for Cosine Similarity?

$\cos(\theta) = (\mathbf{a} \cdot \mathbf{r}) / (|\mathbf{a}| |\mathbf{r}|)$. Because our vectors are unit normalized, it simplifies to the dot product $\sum a_i r_i$.

6. What are the six levels of Bloom's Revised Taxonomy?

Remember $\rightarrow$ Understand $\rightarrow$ Apply $\rightarrow$ Analyze $\rightarrow$ Evaluate $\rightarrow$ Create.

7. What are your exact Bloom transition score thresholds?

Score ≥ 75 : elevate Bloom level. Score $50-74$: maintain level and vary question type. Score < 50 : scaffold downward to preceding level.

8. What are the 5 signals in your scoring rubric and their weights?

Technical Accuracy (30%), Completeness (20%), Relevance (20%), Semantic Similarity (15%), Concept Coverage (15%).

9. Why is semantic similarity alone insufficient for grading?

It measures topical overlap, not correctness. An articulate student asserting that "HTTP runs over UDP" would get a high semantic score without factual accuracy.

10. How does voice input work?

Browser captures audio via Web Audio API; Groq Whisper-large-v3 transcribes it with technical vocabulary prompt biasing in ~ 350 ms.

11. How does voice output work?

Microsoft Edge-TTS streams `en-US-ChristopherNeural` MP3 chunks in-memory via WebSockets directly to the browser.

12. How does the system prevent repeated questions?

Previously asked questions are tracked in `AdaptiveState` and injected into the Groq prompt as negative conditioning constraints.

13. What database stores user data and session histories?

MySQL 8.0 (InnoDB) with 7 relational tables, foreign key cascades, and a `token\_blacklist` table.

14. How do you handle JWT revocation on logout?

When a user logs out, the SHA-256 hash of their JWT is saved in the `token\_blacklist` table. Subsequent requests check this table.

15. How are resumes parsed?

PyMuPDF (`fitz`) extracts raw C-level PDF glyph streams in 12ms and matches against a 50+ canonical CS skill regex taxonomy.

16. What is the difference between Matched Skills and Gap Skills?

Matched Skills = Resume $\cap$ JD. Gap Skills = JD $\setminus$ Resume.

17. What is the purpose of `SKILL\_DOMAIN\_MAP`?

It maps skills to knowledge domains (e.g., SQL $\rightarrow$ DBMS), applying metadata filters in ChromaDB to prevent cross-domain chunk noise.

18. What models run on Groq LPU?

Primary: `openai/gpt-oss-120b` (or `llama-3.3-70b-versatile`). Fallback: `openai/gpt-oss-20b` (or `llama-3.1-8b-instant`).

19. What happens if Groq API goes down?

The system trips an automated failover to the secondary model, and if offline, activates local heuristic scoring via CPU SentenceTransformers.

20. Why did you use 384 dimensions instead of 1,536?

384 dimensions capture all necessary semantic variance for technical sentence matching while executing in <15ms on CPU at zero cost.

21. What is University Syllabus Mode?

An isolated mode where questions are grounded in specific institutional course syllabi (e.g., JNTUH CS501PC) across 5 course units.

22. What does the final Performance Dashboard display?

A 5-axis Radar Chart (Cognitive Depth, Precision, Completeness, Relevance, Breadth), turn-by-turn scores, and a study roadmap.

23. How does FastAPI handle high concurrency?

Using asynchronous coroutines (`async`/`await`) on Uvicorn ASGI, offloading CPU tasks to thread pools.

24. Did AI write this project for you?

No. AI is an accelerator, but software engineers architect systems. The decoupled 4-tier design, Bloom state machine, 5-factor formula, and MySQL schema were engineered line-by-line by our team.

25. What is your Pearson correlation with human interviewers?

$r = 0.89$, proven through our IEEE conference paper ablation study across 100 evaluated turns.

8. EMERGENCY HOSTILE QUESTION DEFENSE CHEAT SHEET

- **If asked: "Why not just use ChatGPT directly?"**

→ Answer: "ChatGPT hallucinates technical standards, drifts off-topic, grades subjectively without anchored rubrics, and has no curriculum alignment or deterministic Bloom state tracking."

- **If asked: "Did you invent the 30/20/20/15/15 weights arbitrarily?"**

→ Answer: "No. They were determined through an empirical ablation study in Table V of our IEEE paper, achieving an $r=0.89$ correlation with human senior engineers and under 3.8% scoring variance."

- **If asked: "What if a candidate attempts a prompt injection jailbreak?"**

→ Answer: "Answers are passed as passive data variables in backticks, and 30% of the score is hardcoded to deterministic cosine similarity (SS) and keyword ratios (KS), dragging malicious answers down."

- **If asked: "Why two databases (MySQL and ChromaDB)?"**

→ Answer: "Polyglot persistence: MySQL provides strict ACID guarantees for relational parent-child entities; ChromaDB provides logarithmic $\mathcal{O}(\log N)$ HNSW vector search."